

EXPERIMENT 30: MORPHOLOGICAL OPERATIONS USING OPENCV – DILATION

Aim

To perform the morphological Dilation operation on an image using Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Convert the grayscale image into a binary image using thresholding.
5. Create a structuring element (kernel).
6. Apply the Dilation operation using `cv2.dilate()`.
7. Display and save the dilated image.

Python Code

```
import cv2
import numpy as np

# Read the input image
img = cv2.imread("input.jpg")

if img is None:
    print("Image not found")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Convert image to binary
_, binary = cv2.threshold(
    gray, 120, 255, cv2.THRESH_BINARY
)
```

```
# Create structuring element
kernel = np.ones((5, 5), np.uint8)

# Apply Dilation
dilated = cv2.dilate(
    binary,
    kernel,
    iterations=1
)

# Save output
cv2.imwrite("dilation_output.jpg", dilated)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Binary Image", binary)
cv2.imshow("Dilated Image", dilated)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the morphological Dilation operation was successfully performed on the input image using OpenCV.